

GALAXIES

Astronomers are drawn naturally to the brightest, the most beautiful, and the most intriguing galaxies. However, of the 100 billion galaxies in the observable universe, only a minority are spectacular spirals and giant ellipticals. Astronomers are beginning to understand that most galaxies are relatively small and faint—diffuse balls and irregular clouds of stars. The faintest and most common

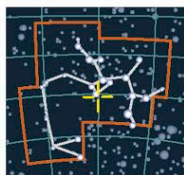
BIG AND BRIGHT

Spirals such as Bode's Galaxy, M81, may be the most attractive type of galaxy, but they are far from the most common. Making up less than 30 percent of all galaxies, they are outnumbered by smaller, fainter galaxies.

galaxies are dwarf ellipticals, which are like oversized globular star clusters of only a few million stars. These feeble galaxies are visible only if they lie nearby in intergalactic terms. The most brilliant are the giant ellipticals, which can be 20 times as luminous as the Milky Way.

DWARF ELLIPTICAL GALAXY

SagDEG



SAGITTARIUS

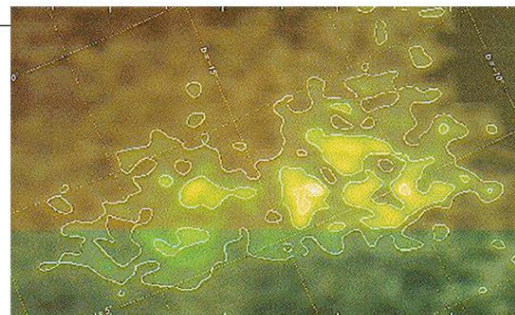
CATALOG NUMBER	None
DISTANCE	88,000 light-years
DIAMETER	10,000 light-years
MAGNITUDE	7.6 for M54 star cluster in SagDEG

The Sagittarius Dwarf Elliptical Galaxy, often called SagDEG, was until recently our closest known

galactic neighbor. It was not found until 1994, and was supplanted only by the discovery of the even closer Canis Major Dwarf in 2003. SagDEG remained hidden for so long because, like all dwarf ellipticals, it is a very faint scattering of stars. It is also well disguised by its position behind the great Sagittarius star clouds that mark our galaxy's center. SagDEG is small and obscure, but it has at least four orbiting globular clusters, which are brighter and more obvious. One of these, M54, was discovered by Charles Messier more than 200 years before the parent galaxy was found.

STAR DENSITY

SagDEG's existence came to light only when a survey of Sagittarius found regions of increased star density—the bright patches in this image.



SagDEG's existence so close to our galaxy is a puzzle. It orbits the Milky Way in less than a billion years and so must have gone through several close encounters that should have ripped it apart and scattered its stars through the galactic

halo. It has survived only due to a large amount of dark matter, producing more gravity than SagDEG's visible stars.

IRREGULAR GALAXY

Large Magellanic Cloud



DORADO

CATALOG NUMBER	None
DISTANCE	179,000 light-years
DIAMETER	20,000 light-years
MAGNITUDE	0.1

The Large Magellanic Cloud (LMC) bears the name of 16th-century explorer Ferdinand Magellan (see panel, opposite). However, cultures native to the Southern Hemisphere have recognized its existence since prehistoric times. Like its smaller

counterpart, the Small Magellanic Cloud, the LMC appears from Earth to be a distinctive, isolated region of the Milky Way, some 10 degrees across, with its own areas of nebulosity and star clusters.

The LMC is in fact an irregular galaxy, orbiting the Milky Way roughly once every 1.5 billion years on a path that brought it to within 120,000 light-years of our galaxy at its closest approach around 250 million years ago. Although the LMC is irregular and is being distorted by the gravity of the Milky Way, it shows some signs of basic structure. Many of its stars are concentrated in a central barlike nucleus, curved at one end. Some astronomers have likened the LMC to a barred spiral with just one arm.

Like all irregular galaxies, the LMC is rich in gas, dust, and young stars, including some of the largest known regions of star birth. One such region is the magnificent Tarantula Nebula, also known as 30 Doradus. It is so brilliant that, if transported to the location of the Orion Nebula (see p.241)—only 1,500 light-years away in the Milky Way—it would be bright enough to cast shadows on Earth at night.

In recent times, the LMC was host to the only bright supernova since the invention of the telescope. Supernova 1987A (see p.266) was observed by astronomers around the world both during and after its explosion, and it has taught astronomers a lot about the final stages of the stellar life cycle.



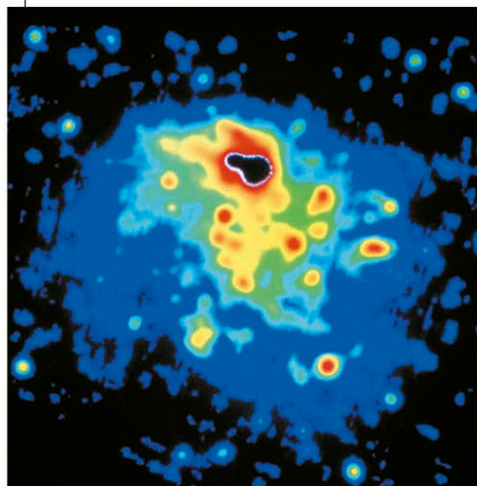
SUPERNOVA BUBBLE

This image shows a bubble of gas around the site of a supernova that exploded about 400 years ago in the LMC. The image is a composite from the Hubble Space Telescope and Chandra X-ray Observatory. Green and blue indicate hot, X-ray-emitting material, and pink shows the visible gas shell shocked by the blast wave from the supernova explosion. The bubble is about 23 light-years across and is expanding at over 11 million mph (18 million kph).



TARANTULA NEBULA

The Tarantula Nebula is home to not one but two huge star clusters. In the center of this image is R136, home to some of the most massive stars known, while at upper right is the slightly older Hodge 301, whose oldest stars have already gone supernova.



RADIO MAP

This false-color radio image of the LMC is centered on the Tarantula Nebula. It shows intense radiation as red and black, indicating ionized hydrogen and star formation.